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Energy efficiency requirements under the EU AI Act

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The rapid proliferation of AI technologies in recent years has led to a growing demand for energy-intensive data centers and high-performance hardware. Laws seeking to regulate AI, such as the EU AI Act, increasingly include rules regarding energy consumption and transparency that will impact businesses developing or using AI technologies.

Calculating AI's global energy consumption precisely is challenging because accurate data on energy consumption is not readily available. Nevertheless, the International Energy Agency estimates that by 2026, the global AI industry will have grown exponentially, and will consume at least ten times the amount of energy that it consumed in 2023. By 2026, electricity consumption by EU data centers is expected to be 30% higher than 2023 levels, as new data facilities are commissioned amid increased AI computations and digitalization.

Against this backdrop, emerging AI regulatory laws are seeking to impose rules regarding energy use in the context of AI. One such law is Regulation (EU) 2024/1689 (the "**EU AI Act**") which entered into force on August 1, 2024 (though enforcement will take effect in stages over several years). The EU AI Act is designed to provide a common set of rules governing the development and use of AI across the EU (and, in some cases, beyond the EU). The EU AI Act aims to ensure "environmental protection, while boosting innovation" and imposes a number of requirements concerning energy consumption and transparency (many of which are yet to be finalized). It is also anticipated that AI regulatory laws in other jurisdictions may include similar requirements regarding energy consumption and transparency. As a result, businesses that develop or use AI are increasingly likely to face energy-related compliance obligations around the world in the coming years.

Transparency requirements for General-Purpose AI Models

The EU AI Act defines General-Purpose AI Models ("**GPAI models**") as AI models that are trained with a large amount of data using self-supervision at scale, that display significant generality, and that are capable of competently performing a wide range of distinct tasks. Examples include large language models such as OpenAI's GPT models, Google's Gemini, and Meta's Llama.

Under the EU AI Act, a business that develops a GPAI model (a "**provider**") is required to create and maintain technical documentation, including a breakdown of the energy consumption of that GPAI model. Where the energy consumption is not yet known, providers of GPAI models may estimate the consumption based on the computational resources used. For clarity, this obligation is limited to the provider of a GPAI model (as distinct from "**deployers**," which are businesses that use GPAI models developed by others).

The AI Office (an EU regulatory / advisory body) is authorized to demand technical documentation on energy consumption from providers with no prior notice. As a result, businesses that develop GPAI models need to keep this documentation regularly updated. Providers of GPAI models launched before August 2, 2025, effectively benefit from a two-year grace period (i.e., they have until August 2, 2027, to demonstrate compliance with the above obligation). However, providers who launch their GPAI models on or after August 2, 2025, will face immediate compliance obligations. This is likely to incentivize providers to launch GPAI models in advance of August 2, 2025.

Systemic risk

The EU AI Act designates certain GPAI models as having "systemic risk" because of their impact on issues such as public health, safety, public security, fundamental rights, or society as a whole. Providers of GPAI models that fall within this category face significant additional compliance obligations, including additional evaluations, assessments, documentation and security requirements.

Energy consumption is one of the factors that can result in a GPAI model being classified as having systemic risk. As a result, providers of GPAI models have an additional incentive to keep energy usage as low as possible, as this may reduce the risk of additional compliance burdens in some cases.

Standards for energy efficiency in AI

The EU AI Act requires the EU Commission to work with existing EU standards bodies and other stakeholders to create standards focused on AI. Among other things, those standards will be aimed at improving resource performance, including energy efficiency, and consumption of other resources during the lifecycle of AI systems and GPAI models.

These EU standards on AI do not yet exist and will likely take some time to come into effect. It is worth noting that the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC) are working on international standards for environmental sustainability in AI, and have produced a draft that is currently in the approval phase. It remains to be seen whether there will be material overlap and consistency between these standards (once finalized) and those to be created under the EU AI Act.

The EU AI Act requires the EU Commission to publish periodic reports on progress on the development of standards for energy-efficient deployment of GPAI models, and the need for further measures or actions. The first of such reports is not due until August 2, 2028. Importantly, these reports may carry recommendations for legally binding corrective measures or actions.

Providers of GPAI models should monitor this space closely, as new standards and recommendations for legally binding measures and actions may have a material impact on their business operations.

Voluntary codes of conduct

The EU AI Act requires regulators to facilitate the creation of voluntary codes of conduct governing, among other things, the impact of AI systems on environmental sustainability, energy-efficient programing, and techniques for the efficient design, training, and use of AI. The EU AI Act requires such voluntary codes of conduct to set out clear objectives and key performance indicators to measure the achievement of those objectives. Again, none of these codes has yet been created, so providers of AI systems and GPAI models should be on the watch for draft codes of conduct as they become available.

Evolving landscape

The EU Commission's focus on energy efficiency in AI continues to evolve. In 2024, it ran a call for tenders (now closed) to measure and foster energy-efficient and low emission AI in the EU. The EU Commission is expected to explore the current and estimated future carbon footprint of AI systems and to develop a framework for measuring compliance with the energy-related objectives of the EU AI Act. It is also exploring a potential AI energy and emissions label, which will presumably operate in a similar manner to existing emissions labelling schemes.

The Energy Efficiency Directive

While the EU AI Act is the first attempt at regulating energy efficiency in AI specifically, it is also worth noting the relevance of the Energy Efficiency Directive ("**EED**"). In keeping with the EU's 2030 target of reducing greenhouse gas emissions by at least 55% (compared to 1990), EU lawmakers revised the EED in 2023 to establish 'Energy Efficiency First' as a fundamental principle of EU energy policy, giving it legal-standing for the first time. The EED requires EU Member States to consider energy efficiency in all relevant policy, planning and major investment decisions taken in the energy and non-energy sectors.

Although the EED does not explicitly mention AI, it emphasizes energy efficiency in the information and communication technology ("**ICT**") sector with a particular focus on data centers. Thus, if a data center accommodates AI, both the EU AI Act and the EED might apply. In this case, a data center operator must comply with the energy efficiency standards and compliance obligations under both pieces of legislation.

The EED allows Member States to include data centers as end consumers of energy in their energy efficiency efforts. Measures may include obligations to consume renewable energy or to reduce the total energy consumption for computing power and other utilities (so-called power usage effectiveness). Additionally, data centers with a total rated energy input exceeding 1 MW must utilize or recover waste heat (so-called energy reuse). Data centers with an average annual energy consumption higher than 85 terajoules (TJ) over the previous three years must implement an energy management system for continuous improvement of the energy efficiency. Transparency and reporting obligations apply.

As a European directive, the EED must be implemented into national law by the Member States by October 11, 2025. For example, Germany has already done so by enacting the Energy Efficiency Act (Energieeffizienzgesetz, "**EnEfG**"), which applies to data centers with a capacity of 300 kW or more. The EnEfG mandates a renewable electricity share of 50%, which increases to 100% from January 1, 2027. Additionally, energy reuse obligations will apply to data centers coming into operation on or after July 1, 2026. Power usage effectiveness obligations must be complied with by July 1, 2027; however, new data centers coming into operation on July 1, 2026, must comply right away. Further transparency and reporting obligations apply.

The EU Taxonomy Assessment Framework for Data Centres

The European Commission's Joint Research Centre published an "Assessment Framework for Data Centres in the Context of Activity 8.1 in the Taxonomy Climate Delegated Act" ("**Assessment Framework**") to facilitate the assessment of data centers under the EU AI Act ("**EU Taxonomy**"). The EU Taxonomy is a classification system that defines criteria for economic activities that are aligned with a net zero trajectory as well as broader environmental goals. The Assessment Framework enshrines rules for classification of data center-related activities with a view to climate change mitigation, building on the voluntary European Code of Conduct for Energy Efficiency in Data Centres.

Conclusion

The EU AI Act is in its infancy, and the majority of its provisions (including those relating to energy consumption and transparency) are not yet in effect. In addition, at the time of writing, not all EU Member States have implemented the EED into their national laws. There is also uncertainty regarding the approaches that national regulators in EU Member States (who are responsible for enforcing many of the requirements of the EU AI Act) will take to the issue of energy efficiency. Non-compliance with the relevant energy-related provisions of the EU AI Act carries potential penalties of up to the greater of €15 million or 3% of worldwide turnover. As a result, businesses that are involved in the development of AI, or that use AI extensively, should closely monitor developments in this space. White & Case's AI Watch regulatory tracker provides continually updated insights into the current state of AI regulation in more than 30 jurisdictions.

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